

Second Terminal Examination - 2082

Class : XI	Subject: Biology (Code: 2011)	Full Marks: 75
Time: 3 hrs.	SET 'A'	Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet. [6 × 1 = 6]

- The border between forest and grassland is an example of
 - Ecotone
 - Ecotype
 - Ecological niche
 - Biome
- Parasitism differs from predation because
 - Predators live inside their prey
 - Parasites do not kill their hosts immediately
 - Predators are always smaller than host
 - Parasites and hosts both benefit
- Ribosomes in a cell are meant for
 - Carbohydrate metabolism
 - Fat synthesis
 - Protein synthesis
 - Osmoregulation
- The storage bank of energy currency of cell is
 - Ribosome
 - Nucleus
 - Plastids
 - Mitochondrion
- The common name of *Marchantia* is
 - Bryophyta
 - Liverwort
 - Dichotomously branched plant
 - Dioecious plant
- A flower with its stalk is called
 - Pedicillate flower
 - Petiolate flower
 - Sessile flower
 - Complete flower

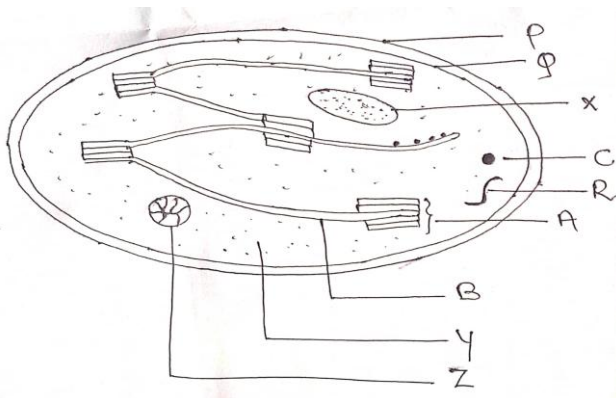
Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

7. Study the following diagram and answer the questions.

[4]



i. Name this diagram.

[0.5]

ii. Why it is called semi- autonomous cell organelle?

[1]

iii. Labell the parts P, Q, A, B and Y.

[0.5×5 = 2.5]

8. Describe the role of temperature on flora & fauna.

[2+2 = 4]

OR

Define binomial system of nomenclature. Write down the rules of binomial system.

[1+3]

9. Why mitochondria are called power house of cell? write down the functions of mitochondria.

[1+3]

10. What is the common name of *Spirogyra*? Draw the scalariform conjugation in *Spirogyra*. (No description)

[1+3 = 4]

Group 'C' [BOTANY]

Give long answer to the following questions.

[8 × 2 = 16]

11. Explain the structural and functional aspects of pond ecosystem.

[8]

12. Describe the family Solanaceae in semi technical terms with floral formula and floral diagram. State the scientific names of any two economic plants of the family.

[3+1+3+1 = 8]

OR

Why bryophytes are called amphibians in plant kingdom? Describe the life cycle of *Marchantia*.

[1+7 = 8]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet. [5 × 1 = 5]

13. Phylogeny represents:
 - a. Embryological history of animal
 - b. Comparative anatomy of animal
 - c. Evolutionary history of animals
 - d. Fossilization of animal
14. The chronological order of human evolution from early to recent is:
 - a. Ramapithecus-H. habilis-Australopithecus-H. erectus
 - b. Ramapithecus- Australopithecus- H. habilis- H. erectus
 - c. Australopithecus- H. habilis- Ramapithecus- H. erectus
 - d. Australopithecus- Ramapithecus- H. habilis- H. erectus
15. Method of food intake in *Paramecium* is
 - a. Holozoic
 - b. Holophytic
 - c. Saprozoic
 - d. Parasitic
16. Shivering characteristic of malaria occurs when
 - a. Schizonts enter the R.B.C.
 - b. The signet ring is formed
 - c. Merozoites are liberated from RBC with toxin
 - d. The schizonts are on the reticuloendothelial cells
17. The female genital aperture in earthworm presents midventrally in the segment:
 - a. 18th
 - b. 14th
 - c. 12th
 - d. 10th

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 =16]

18. Draw a neat and fully labeled diagram of life cycle of *Plasmodium* in man. (Description not required).
19. "Ontogeny recapitulates phylogeny". Elaborate the given statement

OR

Describe the physiology of digestion in earthworm.

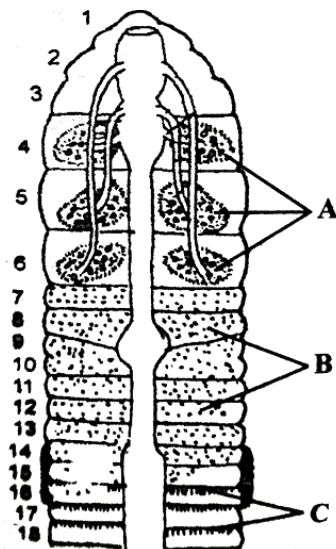
20. Observe the figure given below and answer the questions.

[2+ 1+1]

a. Label A, B and C

b. Which of them are enteronephric and exonephric?

c. Define enteronephric and exonephric.



21. Discuss the autogamy occurring in the *Paramecium caudatum* with well labelled diagram.

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

22. What is conjugation? Describe conjugation in *Paramecium aurelia* with labeled diagrams. Mention any two significances of conjugation. [1+4+2+1]

OR

Name different species of *Plasmodium* causing malaria to human being. Explain the life cycle of *Plasmodium vivax* in mosquito body with labeled diagram. [1+5+3]

23. What causes the emergence of new species? Explain using Modern Synthetic Theory as a foundation. [2+2+4]

ॐ The End ॐ